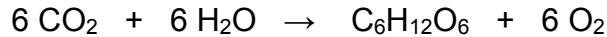


ENERGY FLOW IN FOOD CHAINS

The Sun is constantly putting energy into an ecosystem. Light energy from the sun is used by chlorophyll in plants to split water molecules so that they can react, with carbon dioxide, to form glucose and oxygen. This process is called **photosynthesis** and can be represented by the chemical equation:

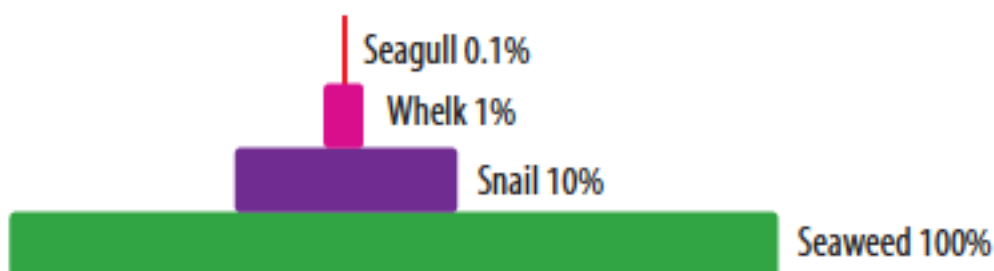


The plants use 90% of the chemical energy (glucose) for everyday living and it is ultimately converted to heat. This is called **respiration** and is the reverse of the photosynthesis equation (this means plants also need oxygen). **The remaining 10% of chemical energy is converted to proteins and other molecules for growth.**

Therefore, **a herbivore that eats the plant only gets the 10% of chemical energy that was originally converted from sunlight.** The herbivore, in turn, uses 90% of the plant's chemical energy for everyday movement, and maintenance converting it to heat. The remaining 10% is used for body growth and development. **This means 10% of the 10% from the plant's chemical energy (i.e. only 1% of the Sun's energy) makes its way into the muscle and other tissues of the herbivore.**

This is decreased even further as we move up the food chain. **A carnivore eats the herbivore.** Ninety per cent of the chemical energy eaten by the carnivore is converted to heat and movement by the animal, leaving just 10% for growth (**0.1% of the original energy absorbed from the Sun by the plant**). This is known as the **10% rule**.

This can be represented diagrammatically in an energy pyramid where producers are the base or first trophic level of the pyramid. The second trophic level is the herbivore and is one-tenth the length of the first trophic level. The third trophic level is the first order carnivore and is one-tenth the size of the second trophic level. There is a limit of 4-5 trophic levels in each energy pyramid as there is not enough energy to be passed on.



An energy pyramid — only about 10 per cent of the food energy received at each level is passed through to the next, the other 90 per cent is transferred to the environment.